Quantification: 536-63x_Hyp_RD_z03c1-2
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PRE-VALIDATION: no number here has been checked against manual (ImageJ) counts.
PIXEL UNITS: lengths px, areas px^2, densities px^-1. The images carry no micron calibration.
Statistics are over interior lacunae only (those not touching the frame edge).

Per image
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measure	value
lacuna count	12
interior lacuna count	12
frame-edge lacuna count	0

Interior lacunae only: mean, median and sample SD (n = 12)
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measure	kind	mean	median	sd	units
area_px2	lacuna shape	2413.42	2332	1414.85	px^2
major_axis_length_px	lacuna shape	93.7767	89.3059	45.516	px
minor_axis_length_px	lacuna shape	36.2309	33.9956	10.5225	px
aspect_ratio	lacuna shape	2.484	2.5713	0.7633	unitless
eccentricity	lacuna shape	0.8772	0.9203	0.0999	unitless
solidity	lacuna shape	0.8024	0.8337	0.1144	unitless
roots_count	headline	9.75	8.5	7.8639	count
ring_length_r30_px	headline	281.833	207	227.901	px
ring_length_r60_px	reported	575.083	513	424.236	px
owned_length_px	ownership-dependent	845.105	470.04	1221.45	px
edge_count	ownership-dependent	40.5	20.5	60.4355	count
mean_edge_length_px	ownership-dependent	22.121	21.1731	7.7186	px
perimeter_px	normalised	290.285	235.622	177.064	px
ring_area_r30_px2	normalised	9790.25	8516	3690.8	px^2
ring_area_r60_px2	normalised	24031.7	22415.5	6739.69	px^2
in_frame_fraction_r30	normalised	0.9777	1	0.0773	unitless
in_frame_fraction_r60	normalised	0.9689	1	0.1021	unitless
ring_density_r30	normalised	0.026593	0.0286601	0.0118828	px^-1
ring_density_r60	normalised	0.0223595	0.0231245	0.0103099	px^-1
roots_per_100px_perimeter	normalised	3.4205	3.4024	1.8559	per 100 px
ring_attached_length_r30_px	network v2	245.417	195	231.79	px
ring_attached_length_r60_px	network v2	395.417	281.5	412.432	px
ring_length_w_r30_px	network v2	319.47	236.359	262.773	px
ring_length_w_r60_px	network v2	649.221	580.387	486.672	px
sholl_crossings_r10	network v2	6.6667	5	5.7419	count
sholl_crossings_r20	network v2	7.1667	6.5	5.0602	count
sholl_crossings_r30	network v2	7.6667	7	5.9289	count
loop_count	loops, report only	2.1667	0	4.6286	count
loop_edge_share	loops, report only	0.0799	0	0.1368	share
ring_width_mean_r30_px	width	7.3477	7.2627	0.7467	px

Per field
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measure	value	units
analysed_area_px2	1.0029e+06	px^2
total_skeleton_length_px	21001	px
canalicular_length_density_per_px	0.0209403	px^-1
junction_count	375	count
junction_density_per_px2	0.00037392	px^-2
lacuna_count	12	count

lacunae_per_px2	1.197e-05	px^-2
skeleton_component_count	392	count
mean_component_length_px	53.574	px
median_component_length_px	19	px
field_length_density_w_per_px	0.0236526	px^-1
field_density_without_flagged_per_px	0.0210848	px^-1
field_density_in_roi_per_px		px^-1
unattached_length_px	904	px
unattached_component_count	48	count
canalicular_length_density_attached_per_px	0.0209403	px^-1
field_length_density_with_unattached_per_px	0.0218417	px^-1
excluded_area_px2	16718	px^2
analysed_field_area_px2	1.03186e+06	px^2
loop_count	35	count
loop_edge_share	0.1132	share
edge_count	1131	count

Canalicular width, the new measure

measure	value	units
width_mean_px	7.5682	px
width_median_px	8	px
width_p10_px	4	px
width_p90_px	10.7703	px
width_pixels_used	20752	count

Canalicular width: 2 x the Euclidean distance transform of the canalicular mask at each skeleton pixel.
Measured only on skeleton pixels on the canalicular mask: the lacuna buffer and the
dilated vascular mask are excluded. The threads are about 2 to 6 px wide, so the width is close to the
pixel grid and quantised in steps of about 1 px, it moves with the threshold (a lower cut widens every
thread), and it has not been validated against any manual measurement.

Lacuna shape

lacuna_id	area_px2	major_axis_length_px	minor_axis_length_px	aspect_ratio	eccentricity	solidity	orientation_rad	centroid_row_px	centroid_col_px	on_border
1	4424	168.508	61.7794	2.7276	0.9304	0.5066	-0.0985	178.037	485.058	False
2	740	35.5513	27.759	1.2807	0.6248	0.9091	-0.3246	161.099	832.037	False
3	3218	123.087	35.0225	3.5145	0.9587	0.8067	-0.0876	285.404	998.443	False
4	4546	152.562	46.2815	3.2964	0.9529	0.6884	-0.3441	401.046	256.605	False
5	3934	133.278	41.5402	3.2084	0.9502	0.7505	-0.3234	452.159	836.983	False
6	2906	117.901	38.3023	3.0782	0.9458	0.7758	-0.3279	473.994	376.288	False
7	2475	101.315	32.9688	3.0731	0.9456	0.8624	-0.3801	520.411	501.561	False
8	2189	73.1101	40.4728	1.8064	0.8328	0.9132	-0.5214	637.506	936.232	False
9	1176	57.7542	28.7935	2.0058	0.8669	0.8276	-0.2231	683.004	402.083	False
10	679	37.9692	24.2376	1.5665	0.7697	0.8573	-0.1349	734.916	174.795	False
11	829	46.9885	25.6067	1.835	0.8385	0.8399	0.1493	873.288	867.089	False
12	1845	77.2965	32.007	2.415	0.9102	0.8909	0.0147	929.902	369.506	False

Network per lacuna

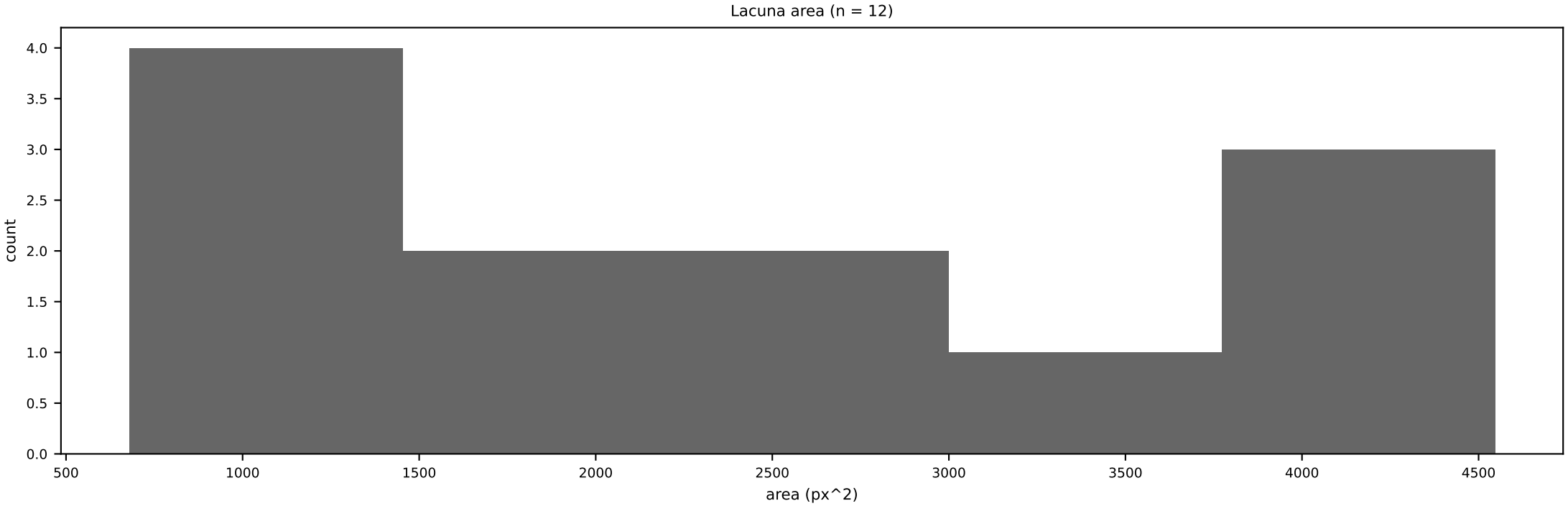
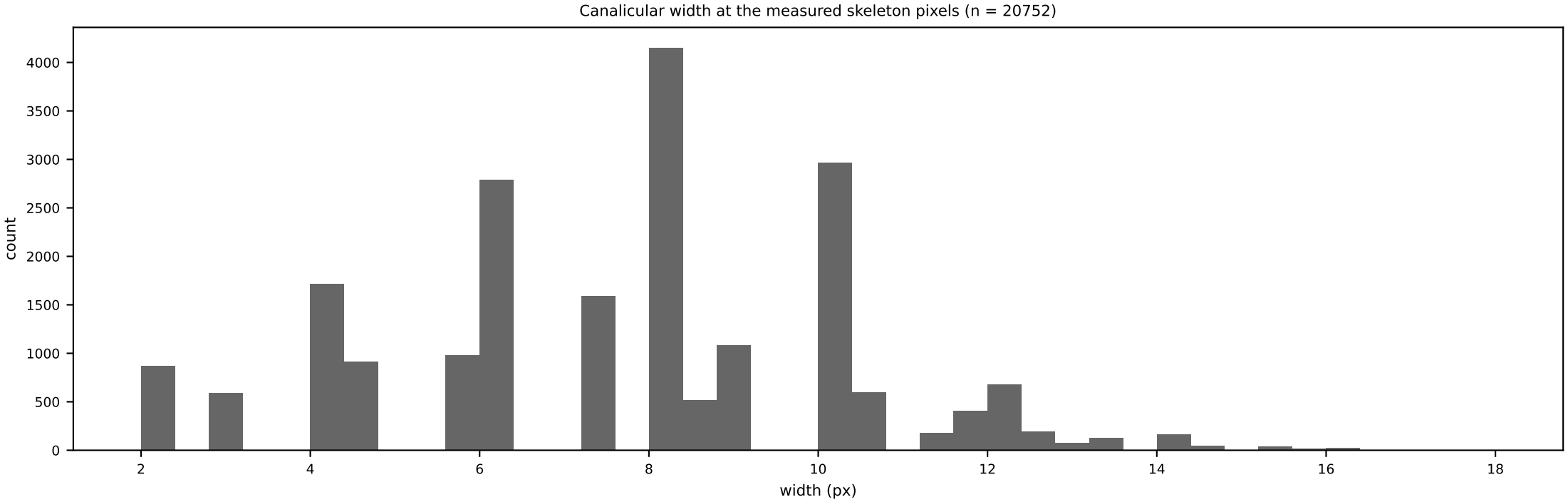
lacuna_id	roots_count	ring_length_r30_px	ring_length_r60_px	owned_length_px	edge_count	mean_edge_length_px	ring_width_mean_r30_px
1	31	912	1720	4284.49	214	20.021	7.2627
2	7	189	377	467.404	21	22.2573	8.8876
3	7	184	421	472.676	20	23.6338	7.1509
4	15	432	905	2214.77	102	21.7134	7.3319
5	13	383	716	707.21	25	28.2884	7.0866

6	11	313	599	309.492	15	20.6328	7.9034
7	9	212	518	348.22	17	20.4835	7.7662
8	2	93	148	26.8995	3	8.9665	
9	9	273	543	479.86	31	15.4794	7.208
10	8	202	508	486.475	22	22.1125	7.5785
11	1	44	86	41.7279	1	41.7279	5.8562
12	4	145	360	302.037	15	20.1358	6.7922

Normalised per lacuna

lacuna_id	perimeter_px	ring_area_r30_px2	ring_area_r60_px2	in_frame_fraction_r30	in_frame_fraction_r60	ring_density_r30	ring_density_r60	roots_per_100px_perimeter
1	733.068	18106	39240	1	1	0.05037	0.0438328	4.2288
2	114.811	5937	17499	1	1	0.0318343	0.0215441	6.097
3	328.049	8327	18178	0.7322	0.645	0.0220968	0.0231599	2.1338
4	472.659	14091	31679	1	1	0.0306579	0.0285678	3.1735
5	380.534	12648	30304	1	1	0.0302815	0.0236272	3.4163
6	324.635	11576	25943	1	1	0.0270387	0.0230891	3.3884
7	246.551	9801	24158	1	1	0.0216304	0.0214422	3.6504
8	224.693	8620	22726	1	0.995	0.0107889	0.00651236	0.8901
9	188.93	7444	20339	1	1	0.0366738	0.0266975	4.7637
10	120.125	5980	17550	1	1	0.0337793	0.0289459	6.6597
11	144.368	6541	18659	1	1	0.0067268	0.00460904	0.6927
12	204.995	8412	22105	1	0.9866	0.0172373	0.0162859	1.9513

Network v2 per lacuna

lacuna_id	ring_attached_length_r30_px	ring_attached_length_r60_px	ring_length_w_r30_px	ring_length_w_r60_px	sholl_crossings_r10	sholl_crossings_r20	sholl_crossings_r30
1	890	1584	1045.86	1964.29	21	19	24
2	120	178	204.225	411.764	3	6	4
3	180	291	201.368	466.291	5	4	5
4	406	624	485.304	1014.58	13	12	12
5	358	543	435.889	806.796	10	9	8
6	215	272	364.676	685.269	5	10	9
7	210	338	238.823	587.943	6	7	7
8	30	30	97.598	158.711	1	1	2
9	233	371	315.907	635.441	8	8	7
10	123	247	233.894	572.831	3	6	8
11	40	49	47.799	90.598	1	1	2
12	140	218	162.297	396.149	4	3	4

536-63x_Hyp_RD_z03c1-2: distributions, pre-validation, px